

Week 6 Review Session

PH142 Team - Summer 2026

Inference for proportion

Conditions for Inference about a Proportion

- Data are a simple random sample from the population
- The sample size n is large enough to ensure that the sampling distribution of $\hat{p}$ is close to normal
- (sampling distribution for $\hat{p}$)
Centered on p with a standard error of $\sqrt{\frac{p(1-p)}{n}}$

Four Types of CIs for a Proportion

The methods give similar results when the Normal approximation assumptions are satisfied.

- **Large Sample: low coverage**

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

The large sample method is easy to calculate — you only need $\hat{p}$ and n . However, if we do not have a sampling distribution that approaches Normal, this confidence interval does not perform well, meaning that even if you think you should have 95% confidence that the CI contains the true value p , it is often much lower.

Four Types of CIs for a Proportion

The methods give similar results when the Normal approximation assumptions are satisfied.

- **Plus Four:** Use when $n \geq 10$ and confidence level $\geq 90\%$

$$\tilde{p} \pm z^* \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{n + 4}} \quad \tilde{p} = \frac{\text{number of successes} + 2}{n + 4}$$

An easy way to save the CI. If you add 2 imaginary successes and 2 failures to the dataset (increasing the sample size by 4 imaginary trials), the interval performs well again.

Four Types of CIs for a Proportion

The methods give similar results when the Normal approximation assumptions are satisfied.

- **Wilson Score: In R, `prop.test()`**
 - Example: `prop.test(x = 100, n = 500, conf.level = 0.95)`
 - The default with the `prop.test` function is two-sided, but you can change this by specifying “less” or “greater.”
- **Exact (Clopper Pearson): In R, `binom.test()`**
 - Example: `binom.test(x = 100, n = 500, conf.level = 0.95)`
 - It's statistically conservative, which means that it gives better coverage than it suggests. So a 95% CI computed under this method includes the true proportion in the interval $> 95\%$ of the time.

Hypothesis Testing for a Proportion

You're interested in knowing whether there is evidence against the null hypothesis that the population proportion $\mathbf{p}$ is equal to some specified value $\mathbf{p_0}$. That is $\mathbf{p=p_0}$.

$$H_0 : p = p_0$$

$$H_a : p > p_0 \text{ or } p < p_0 \text{ or } p \neq p_0$$

- Use this test when $np_0 \geq 10$ and $n(1-p_0) \geq 10$

- Test statistic:
$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Sample Size for a Proportion

Suppose that you want to estimate a sample size for a proportion within a given margin of error. That is, you want to put a maximum bound on the width of the corresponding confidence interval.

$$n = \left(\frac{z^*}{m}\right)^2 p^*(1 - p^*)$$

- m denote the desired margin of error
- p^* is your best estimate for the underlying proportion. If you have no best guess, $p^* = 0.5$ will produce the most conservative estimate of n

Example

You conducted a survey of randomly chosen women from California and found that 775 out of 5615 new mothers exclusively breastfed their infants. Suppose that in 2010, the rate of exclusive breastfeeding in California was known to be 18.6%. Is your rate different from 18.6%?

- **Part 1: Hypothesis Test**

- 1. Define your null and alternative hypotheses.
- 2. Calculate $\hat{p}$
- 3. Calculate SE
- 4. Calculate the z-test for the proportion
- 5. Calculate and interpret the p-value using an R function

- **Part 2: Sample Size Calculation**

- Suppose you are a public health researcher planning a new study for next year to re-estimate the rate of exclusive breastfeeding. How large a sample would you need to estimate the proportion with a margin of error (m) of $\pm 3\%$ at a 95% confidence level?

Answers

- $H_0: p = 0.186, H_A: p \neq 0.186$
- $\hat{p} = 775/5615 = 0.1380232$
- $SE = \sqrt{p_0(1-p_0) / n} = 0.186(1-0.186) / 5615 = 0.00519$
- $Z = (\hat{p} - p_0) / SE = (0.1380232 - 0.186) / 0.00519 = -9.239$
- $\text{pnorm}(q = -9.239) * 2$

```
pnorm(q = -9.239266)*2
```

```
## [1] 2.481939e-20
```

- $n \approx 646.29 \quad n = \left(\frac{1.96}{0.03}\right)^2 \times 0.186 \times (1 - 0.186)$

CI for the Difference of Two Proportions

- Large Sample
 - Low coverage
 - Use when the number of observed successes and failures are > 10 for both samples

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

CI for the Difference of Two Proportions

- **Plus Four**

- When the assumptions of the large sample method are not satisfied, we use the plus four method.
- Use when the sample size is ≥ 5 , with any counts of success and failure
- Much more accurate when the sample sizes are small
- May be conservative (giving a higher level of confidence than the one specified)

$$\tilde{p}_1 = \frac{\text{no. of successes in pop1} + 1}{n_1 + 2}$$

$$\tilde{p}_2 = \frac{\text{no. of successes in pop2} + 1}{n_2 + 2}$$

$$(\tilde{p}_1 - \tilde{p}_2) \pm z^* \sqrt{\frac{\tilde{p}_1(1-\tilde{p}_1)}{n_1+2} + \frac{\tilde{p}_2(1-\tilde{p}_2)}{n_2+2}}$$

Hypothesis Testing for the Difference of Two Proportions

$$H_0 : p_1 = p_2 \text{ or } p_1 - p_2 = 0$$

$$H_a :$$

- $p_1 \neq p_2$ or $p_1 - p_2 \neq 0$ (two-sided)
- $p_1 > p_2$ or $p_1 - p_2 > 0$ (one-sided upper tail)
- $p_1 < p_2$ or $p_1 - p_2 < 0$ (one-sided lower tail)
- Best estimate of the underlying proportion: $\hat{p} = \frac{\text{no. successes in both samples}}{\text{no. individuals in both samples}}$
- Best guess of the SE for $\hat{p}$ under this null hypothesis: $\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$

z-test for the Difference Between Two Proportions

- Use this test when the counts of successes and failures are ≥ 5 in both samples

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Chi-Squared

The Chi-Square Test Statistic

- What do we do with a categorical variable when there are more than 2 categories?
- The Chi-Square Test Statistic

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

- Quantifies the magnitude of the difference between observed and expected counts under the null hypothesis
- k is the number of cells in the table.
- O_i is the observed count for the i th group
- E_i is the expected count for the i th group

Chi-Square Goodness of Fit

- How good do the expected counts “fit” the observed counts?
- Test with discrete outcome measured in a single sample, where the goal is to assess whether the distribution of responses follows a known distribution

H_0 : proportion in observed is the same as the proportion from a given population

$$p_1 = \#_1, p_2 = \#_2, \dots, p_k = \#_k$$

where $\#_1, \#_2, \dots, \#_k$ are provided or can be derived from percentages provided

H_a : At least one of p_k is not equal to the proportion stated in the null hypothesis

- Degrees of freedom = $k - 1$
- In R: `pchisq(q, df, lower.tail = F)`

Conditions for Chi-Square Goodness of Fit

- Fixed n of observations
- All observations are independent of one another
- Each observation falls into just one of the k mutually exclusive categories
- The probability of a given outcome is the same for each observation
- All expected values greater than 1 ($E_i > 1$)
- At least 80% of the cells have 5 or more observations ($O_i \geq 5$ for $\geq 80\%$ of the cells)

Chi-Square Test of Independence

- Investigate the relationship between two categorical variables
- Goal of the analysis is to compare the distribution of responses to the discrete outcome variable among several independent comparison groups

H_0 : Response and explanatory variables are independent

H_a : Response and explanatory variables are dependent

- Degrees of freedom = $(r - 1)(c - 1)$
- In R: `pchisq(q, df, lower.tail = F)`

Conditions for Chi-Square Test of Independence

- $E_i \geq 5$ for at least 80% of the cells
- All $E_i > 1$
- If table is 2x2, all four cells need $E_i \geq 5$
- Must have data from:
 - Independent SRSs from ≥ 2 populations, with each individual classified according to one category (i.e., each individual can only belong to one cell in the table so the categories need to be mutually exclusive)
 - A single SRS, with each individual classified according to each of two categorical variables

Example

A randomized trial is designed to evaluate the effectiveness of a newly developed pain reliever designed to reduce pain in patients following joint replacement surgery. The trial compares the new pain reliever to the pain reliever currently in use (called the standard of care). 100 patients are enrolled and assigned to either the new pain reliever or standard. They indicated pain levels 0-10 before and after surgery. The primary outcome was a reduction in pain of 3 or more scale points (defined by clinicians as a clinically meaningful reduction). We want to test whether there was a significant difference in the proportions of patients reporting a meaningful reduction (i.e., a reduction of 3 or more scale points)

Treatment Group	# with Reduction of 3+ Points	# with Reduction of <3 Points	Total
New Pain Reliever	23 (17.0)	27 (33.0)	50
Standard Pain Reliever	11 (17.0)	39 (33.0)	50
Total	34	66	100

Example

- Set up hypotheses and determine level of significance
- Compute the test statistic.
- Plug statistic and degrees of freedom into `pchisq()` in R to get p-value.

Treatment Group	# with Reduction of 3+ Points	# with Reduction of <3 Points	Total
New Pain Reliever	23 (17.0)	27 (33.0)	50
Standard Pain Reliever	11 (17.0)	39 (33.0)	50
Total	34	66	100

Answers

- H0: Treatment and outcome (meaningful reduction in pain) are independent; Ha: Treatment and outcome are not independent

- $$\chi^2 = \frac{(23-17.0)^2}{17.0} + \frac{(27-33.0)^2}{33.0} + \frac{(11-17.0)^2}{17.0} + \frac{(39-33.0)^2}{33.0}$$
$$\chi^2 = 2.12 + 1.09 + 2.12 + 1.09 = 6.42$$

- $df = (2-1)*(2-1) = 1$
- ```
> pchisq(q = 6.42, df = 1, lower.tail = F)
```

```
[1] 0.01128422
```

- The p-value is less than 0.05, so we reject the null hypothesis that treatment and outcome are independent

# Chi-Square Test of Independence in R

- Example from Lecture 26 Slide 34-35

```
library(tibble)
two_way <- tribble(~ sandwich, ~ nosandwich,
 109, 4, #row for Illness
 116, 34) #row for no Illness

chisq.test(two_way, correct = F) #not using Yates' correction

##
Pearson's Chi-squared test
##
data: two_way
X-squared = 19.074, df = 1, p-value = 1.257e-05
```

- To compute the chi-square test in R, we need to first put this two-way table into a data frame.
- Then, we use `chisq.test()`. We set `correct=F` (when  $n < 100$  or any  $O_i < 10$ ) to get a value closer to what we calculated by hand - there will be some differences here because of rounding.



# Relationship Between the Chi-Square Test and the Two-Sample Z-Test

- The z-test comparing two proportions is equivalent to the chi-square test of independence
- The p-value from the z-test for two proportions is equal to the p-value from the chi-square test, and the z-statistic is equal to the square root of the chi-square statistic in this situation.

# What we covered in this class?



- **Foundations & Descriptive Statistics**

- Introduction to PPDAC & Visualization
- Summarizing data numerically (central tendency & spread)
- Exploring relationships between two variables
- Introduction to Regression
- Two-way tables (categorical relationships)
- Samples, observational studies & experimental design

- **Probability & Distributions**

- Introduction to Probability
- The Normal Distribution
- The Binomial Distribution
- Poisson Distribution
- Sampling and Central Limit Theorem

- **Statistical Inference**

- Confidence Intervals (known SD)
- Hypothesis Testing basics
- Power, Type I & II errors, sample size
- Inference about population mean (z and t tests)
- Two-sample and paired t-tests
- ANOVA, Tukey, Bonferroni
- Non-parametric tests
- Inference for regression

- **Categorical Data Analysis**

- Inference for proportions
- Four types of CIs for proportions
- Hypothesis testing for proportions
- Difference between two proportions
- Chi-squared goodness of fit
- Chi-squared test of independence
- Two-way tables analysis